

Section 51.

MathScienc

#4.  $\text{irr}(i, \mathbb{Q}) = x^2 + 1 (= (x+i)(x-i) \text{ in } \overline{\mathbb{Q}})$

$\text{irr}(\sqrt[3]{2}, \mathbb{Q}) = x^3 - 2 (= \prod_{k=0,1,2} (x - \sqrt[3]{2} \cdot \exp(2\pi i \cdot \frac{k}{3}) \text{ in } \overline{\mathbb{Q}})$

Take  $\alpha \in \mathbb{Q}$  satisfying

$\alpha \neq \frac{\sqrt[3]{2}}{2i} \cdot (-1 + \exp(2\pi i \cdot \frac{k}{3})) \quad (k=1,2), \frac{\sqrt[3]{2}}{2}$

$= \frac{\sqrt[3]{2}}{2i} (-\frac{3}{2} + \frac{\sqrt{3}i}{2}) \text{ or } \frac{\sqrt[3]{2}}{2i} (-\frac{3}{2} - \frac{\sqrt{3}i}{2})$

$\frac{\sqrt{2}, \sqrt{3}}{\sqrt{2} + \sqrt{3}}$

Let  $\alpha = 1$ , and  $\alpha = \sqrt[3]{2} + i$ . Denote  $\sqrt[3]{2} = \zeta$ . ( $\zeta^3 = 2$ , clearly)

$\alpha^2 = \zeta^2 - 1 + 2i\zeta = (-1, 0, 1, 0, 2, 0)$

$\alpha^3 = 2 - i + 3\zeta^2\zeta - 3\zeta = (2, -3, 0, -1, 0, 3) \quad \alpha^4 = (1, \zeta, \zeta^2, i, i\zeta, i\zeta^2)$

$\alpha^4 = 2\zeta - i\zeta + 6i - 3\zeta^2 + 2i + (-3\zeta^2 - 3\zeta i) = (1, 2, -6, 8, -4, 0)$

$\alpha^5 = (-8, 4, 0, 1, 2, -6) + (-12, 1, 2, 0, 8, -4) = (-20, 5, 2, 1, 10, -10)$

(Since  $[\mathbb{Q}(\sqrt[3]{2}, i) : \mathbb{Q}] = 6$ ,  $\alpha^n$  ( $n \geq 6$ ) are meaningless).

The set  $\{\alpha^n \mid 0 \leq n \leq 5\}$  is linearly independent, therefore, we can find two polynomials in  $\alpha$  as

$i = (a_0, 0, 0, 1, 0, 0) = \sum_{n=0}^5 a_n \alpha^n$

where  $a_n, b_n \in F$ .  $n^3$

$\sqrt[3]{2} = (0, 1, 0, 0, 0, 0) = \sum_{n=0}^5 b_n \alpha^n$

Take  $A = \begin{pmatrix} 1 & [\alpha]_F & [\alpha^2]_F & \dots & [\alpha^5]_F \end{pmatrix} \in M_{6 \times 6}(F)$

$A \begin{pmatrix} a_0 \\ \vdots \\ a_5 \end{pmatrix} = i = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{pmatrix} a_0 \\ \vdots \\ a_5 \end{pmatrix} = A^{-1} \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$  (Specifically,  $(a_0, \dots, a_5) = \frac{1}{22} \cdot (9, 100, 18, 40, -9, 12)$ )

and

$A \begin{pmatrix} b_0 \\ \vdots \\ b_5 \end{pmatrix} = \zeta = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{pmatrix} b_0 \\ \vdots \\ b_5 \end{pmatrix} = A^{-1} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

Finally, calculating  $A^{-1}$ , we obtain the result.

$\hookrightarrow$

Alternatively, take  $\alpha = \sqrt[3]{2} \cdot i$ . Then,  $\sqrt[3]{2} = \alpha^4 \cdot \frac{1}{2}$  and  $i = -2\alpha^3$ , thus

$F(\alpha) = F(\sqrt[3]{2}, i)$

$\underbrace{\hspace{2cm}}$

$\underbrace{\hspace{2cm}}$

#11. Algebraic extension of perfect is perfect.

$K$   
 $\downarrow$   
 $E$   
 $\downarrow$   
 $F$

Proof. Let  $F$  be a perfect field and  $E/F$  be an algebraic extension.

Let  $K/E$  be a finite extension of  $E$ .

Suppose that  $K$  is not separable over  $E$ . Then,

$$\{[K:E]\} < [K:E].$$

This implies that for some  $\alpha \in K$ ,  $\text{irr}(\alpha, E)$  has common multiplicity larger than 1.

Meanwhile, since  $K/E$  finite and  $E/F$  algebraic,  $\alpha \in K$  is algebraic over  $F$ .

But, since  $F$  is a perfect field,  $\text{irr}(\alpha, F)$  is separable over  $F$ .

But!!  $\text{irr}(\alpha, E)$  divides  $\text{irr}(\alpha, F)$ , it is contradiction.

#12. Let  $F \subseteq E \subseteq K$ ,  $E/F$  algebraic,  $K/E$  algebraic.

Let  $\alpha \in K$  be given. Since  $\alpha$  is algebraic over  $E$  and  $E/F$  algebraic,  $\alpha$  is algebraic over  $F$ . Therefore,  $\text{irr}(\alpha, F)$  exists.

By the assumption,  $\text{irr}(\alpha, E)$  have common multiplicity as 1. Hence

$$\text{irr}(\alpha, E) = \prod a_n x^n \quad \text{where } a_n \in E.$$

Since  $\text{irr}(\alpha, E)$  is contained in  $F(a_0, \dots, a_{\deg \alpha})$  with multiplicity 1,

$\alpha$  is separable over  $F(a_0, \dots, a_{\deg \alpha})$ , which implies that

$\alpha$  is separable over  $F$ . ( $F(a_0, \dots, a_{\deg \alpha})$  is separable, by the assumption)

#13. Let  $F_s = \{\alpha \in E \mid \alpha \text{ is separable over } F\}$

Clearly  $F \subseteq F_s$ . Let  $\alpha, \beta \in F_s$  be given.

Since  $F(\alpha + \beta) \subseteq F(\alpha, \beta)$  and  $F(\alpha, \beta)$  is separable over  $F$ ,  $F(\alpha + \beta)$  is separable.

Similarly  $F(\alpha\beta)$  and  $F(\alpha/\beta)$  ( $\beta \neq 0$ ) too.

#14.

a). For any  $\alpha \in E$ ,  $\sigma_p^n(\alpha) = \alpha^{p^n} = \alpha$ , because the finite field of order  $p^n$  is set of zeros of  $x^{p^n} - x$ .  
If order of  $\sigma_p$  is smaller than  $n$ , then it is contradiction to the above fact.

b). Since  $E$  is finite field,  $E/\mathbb{Z}_p$  separable and simple extension.

And  $E$  is splitting field of  $\mathbb{Z}_p$ , because it is generated by all roots of  $x^{p^n} - 1$ .

Now,  $|G(E/\mathbb{Z}_p)| = [E:\mathbb{Z}_p] = n$  with

$\sigma_p \in G(E/\mathbb{Z}_p)$  has order of  $n$ ,

$$G(E/\mathbb{Z}_p) = \langle \sigma_p \rangle.$$

$$\sqrt[3]{2}, \sqrt[3]{2} \cdot \frac{-1 + i\sqrt{3}}{2}, \sqrt[3]{2} \cdot \frac{-1 - i\sqrt{3}}{2}$$

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#4. Clearly,  $[K:\mathbb{Q}(\sqrt[3]{2}, i)] = 2$ . Thus  $|G(\mathbb{Q}(\sqrt[3]{2}, i))| = 2$ .

#7. Similarly,  $[K:\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})] = 2$ , gives that  $|G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3}))| = 2$ .

(Since  $[\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3}):\mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3}):\mathbb{Q}(\sqrt[3]{2})] \cdot [\mathbb{Q}(\sqrt[3]{2}):\mathbb{Q}] = 4$ ,

$$[K:\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})] = 2.)$$

#11. Find  $\text{Gal}(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q})$ . Denote  $\zeta_k = \sqrt[3]{2} \cdot \exp(2\pi i \frac{k}{3})$   $k=1, 2$

a).  $\sigma_0 = \text{identity}$ .  $\sigma_1$   $\sqrt[3]{2}$  maps to  $\zeta_1$  and  $i\sqrt{3}$  maps to  $i\sqrt{3}$  3  
 $\sigma_2$  " to  $\zeta_2$  and " 2  
 $\sigma_3$  " to  $\sqrt[3]{2}$  and  $i\sqrt{3}$  maps to  $-i\sqrt{3}$  2  
 $\sigma_4$  " to  $\zeta_1$  and " 3  
 $\sigma_5$  " to  $\zeta_2$  and " 2

b). By the Cauchy's Theorem, group of order 6 is  $\mathbb{Z}_6$  if it is Abelian, and  $S_3$  if it is non-Abelian.

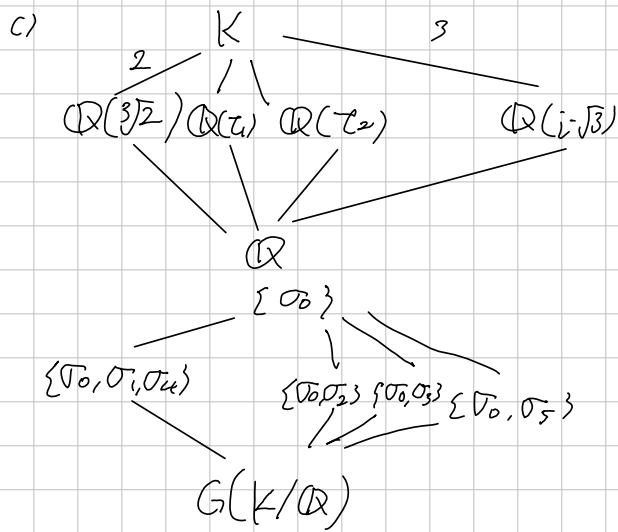
Enough to show  $G(K/\mathbb{Q})$  is not Abelian. Observe that

$$(\sigma_1 \circ \sigma_3)(\sqrt[3]{2}) = \sigma_1(\sqrt[3]{2}) = \zeta_1 \text{ but}$$

$$(\sigma_3 \circ \sigma_1)(\sqrt[3]{2}) = \sigma_3(\zeta_1) = \sqrt[3]{2} \cdot (-1 - i\sqrt{3})/2 \neq \zeta_1. \text{ Thus } G(K/\mathbb{Q}) \text{ is not Abelian.}$$

$$\sigma_1^2(\sqrt[3]{2}) = \sigma_1(\zeta_1) = \zeta_1.$$

$$\sigma_5^2(\sqrt[3]{2}) = \sigma_5(\zeta_2) =$$



#6. Since  $G(K/E) \trianglelefteq G(K/F)$ ,  

$$G(E/F) \cong G(K/F) / G(K/E)$$
 and  $G(K/F)$  is abelian  $\Rightarrow$  the  $G(E/F)$  is abelian.  
 $G(K/E)$  is abelian, because it is subgroup of abelian.

Thm. Let  $K$  be a finite normal extension of  $F$ , and  $F \leq E \leq K$ .

Then  $K$  is a finite normal extension of  $E$  and

$$\text{Aut}(K/E) \leq \text{Aut}(K/F).$$

$$\begin{array}{c} K \\ | \\ E \\ | \\ F \end{array}$$

Moreover, two Automorphisms  $\sigma$  and  $\tau$  in  $\text{Aut}(K/F)$  are satisfied  $\sigma|_E = \tau|_E$

if and only if

$$\sigma \text{Aut}(K/E) = \tau \text{Aut}(K/E)$$

Proof. Above two assertions are trivial. Let  $\sigma, \tau \in \text{Aut}(K/F)$  be given.

$$\sigma \text{Aut}(K/E) = \tau \text{Aut}(K/E) \Leftrightarrow \tau^{-1}\sigma = \mu \text{ for some } \mu \in \text{Aut}(K/E)$$

$$\Leftrightarrow \sigma|_E = \tau|_E$$

Def. If  $K$  is a finite Normal Extension of  $F$ , then  $\text{Aut}(K/F)$  is called

Galois Group of  $K$  over  $F$ .

Galois Theory.

Let  $K/F$  be a finite normal extension.

$$\lambda: \{E | F \leq E \leq K\} \rightarrow \text{Gal}(K/F): E \mapsto \text{Gal}(K/E)$$

For any  $F \leq E \leq K$ ,  $[K:E] = |\text{Gal}(K/E)|$ ,

$$[E:F] = |\text{Gal}(K/F) : \text{Gal}(K/E)|$$

$E$  is Normal of  $F \Leftrightarrow G(K/E)$  is Normal subgroup of  $G(K/F)$ .

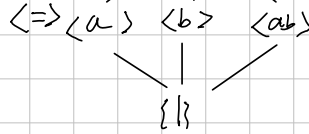
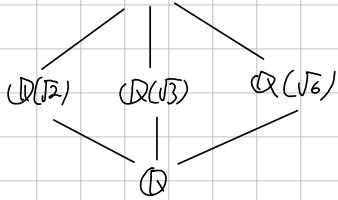
If  $E$  is Normal of  $F$ , then

$$G(E/F) \cong G(K/F)/G(K/E)$$

$$\mathbb{Q}(\sqrt{2}, \sqrt[3]{2})$$

$$\sqrt{2} \cdot \sqrt[3]{2}$$

$$\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2+\sqrt{3}})$$



$$\text{irr}(\sqrt{2}, \mathbb{Q})$$

$$\text{irr}(\sqrt[3]{2}, \mathbb{Q})$$

$$\alpha = \sqrt{2} + \sqrt{3}$$

$$\alpha^2 = 5 + 2\sqrt{6}$$

$$\alpha^3 = 5\sqrt{2} + 4\sqrt{3} + 5\sqrt{3} + 6\sqrt{2} = 11\sqrt{2} + 9\sqrt{3}$$

$$(\alpha^3 - 11\alpha) \cdot -\frac{1}{2} = \sqrt{3}$$

$$(\alpha^3 - 9\alpha) \cdot \frac{1}{2} = \sqrt{2}$$

$$(a_1 + \dots + a_m)^n$$

$$= (a_1 + (a_2 + \dots + a_m))^n$$

$$= \sum \binom{n}{k_1} a_1^{n-k_1} \cdot (\dots)^{k_1}$$